

The kanban approach, between agility and leanness: a systematic review

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Abstract The interest in lean product development in general and the Kanban approach in particular has increased over the years. However, practitioners, in the software development field, have significant challenges in implementing the Kanban approach as it lacks a clear definition of its principles, practices, techniques and tools. This study aims to provide insight into the Kanban approach and its elements (concepts, principles, practices, techniques, and tools) that have been empirically reported by scholars and practitioners. This insight is produced by using the systematic review method to analyze the available literature. A total of 37 primary studies were selected from more than 3,000 unique studies. Our findings show that the primary studies have considered and reported 20 different elements as part of the Kanban approach based upon considerations of being an agile approach or a lean principle; these elements have realized great benefits and improvements to the software development teams. These benefits along with the challenges have been reported in this study. Due to the variety of organization types, contexts, and project sizes reported in the primary studies, it is expected that the results in this study would help in establishing knowledge on what are the different elements of the Kanban approach as well as offering a first step towards developing guidelines for practitioners to help in introducing the Kanban approach to software development organizations.

Keywords Kanban · Lean · Agile · Software development · Software engineering · Process improvement · Systematic review

1 Introduction

In recent years, the lean approach to software development and its concepts have become increasingly popular. The lean approach was first applied in the manufacturing industry. It was devised at Toyota and was originally called the Toyota Production System (TPS). The aim of

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the lean approach is to deliver value to customers more effectively and efficiently through the process of finding and eliminating waste, which is a huge impediment to the productivity and quality offered by an organization (Liker and Hoseus 2008; Magee 2008).

One of the most popular principles of the lean approach is kanban, which is a tool for controlling the logistical chain from a production point of view and is a method by which just-in-time (JIT) is achieved (Ohno 1988). Kanban is one of the two pillars in the lean house that was developed by Toyota. Since 2003, David Anderson (2010) has attempted to tailor the Kanban system to software development, formulating the *Kanban method*, which applies incremental, evolutionary process and systems changes in organizations.

Anderson differentiates the *Kanban method* from kanban—the pull system—by capitalizing the word Kanban. He identifies five elements to the successful implementation of the Kanban method: 1) visualize the workflow, 2) limit work-in-progress (WIP), 3) manage flow, 4) make policies explicit, and 5) implement feedback loops. Anderson's contribution to establishing a comprehensive Kanban approach is evident; however, the approach is not based upon a well-defined framework and therefore needs further development.

The purpose of this research study is to illuminate the guiding principles and elements of the Kanban approach. The objective is to increase the likelihood of implementing the Kanban approach successfully to software product development by 1) Defining the prime elements of the Kanban approach, and 2) Outlining the benefits and drawbacks of using the Kanban approach to help IT organizations as they decide whether to apply the approach.

The findings of this systematic literature review were derived from 37 primary studies that were selected per the research scope. An analysis of these studies showed that they can be classified into four main categories: 1) Kanban as an element of lean (beyond agile), 2) Kanban as an agile methodology, 3) Hybrid of Kanban and Scrum, and 4) Explanatory Kanban. Each classification identifies several sets of elements related to the Kanban approach that scholars in different categories—and sometimes within the same category—have adopted. The existence of multiple sets of elements added another dimension to the difficulty of establishing a common understanding and a clear definition of the Kanban approach among practitioners.

Furthermore, based on the findings of this review, the investigators propose areas that require further analysis and investigations. This will ensure that software development organizations are well informed regarding all that is involved in applying the Kanban approach to their software development processes.

The paper is organized as follows: Section 2 provides background related to the research area and emphasizes the differences between this systematic review and previous systematic reviews in the domain. Section 3 outlines the research methodology by way of a systematic review. The results of the study are given in Section 4, together with their analysis. Section 5 presents a discussion of the study, including proposed guidelines to implement the Kanban approach and threats to the study's validity, while Section 6 covers conclusions as well as recommendations for extending the research study.

2 Problem Statement and Scope

The value that the lean approach has brought to the manufacturing industries has inspired scholars and practitioners in the software engineering domain to begin exploring and implementing the lean concepts and principles to the software development process. In 2003, Poppendieck and Poppendieck (2003) wrote the first book on lean principles to software development. They proposed key elements of lean principles and outlined the challenges to

implement the lean approach successfully in IT organizations. The main challenge has been related to the underlying changes to the organization culture that the lean thinking brings to an organization.

Poppendieck and Poppendieck (2003), on the other hand, claimed that understanding lean thinking has helped IT organizations to achieve sustainable performance improvement. However, Shah and Ward (2007) questioned the level of understanding possessed by scholars of lean thinking in the software engineering domain. The hypothesis is that practitioners are usually simulating practices of proven approaches from other disciplines or industries to software development, where as these practices are typically context-dependent and should be different from one setting to another. Instead, practitioners and scholars should understand the underlying principles in order to develop best practices around the application strategies of these principles (Poppendieck and Poppendieck 2003; Wang et al. 2012; Al-Baik and Miller 2014).

Although the Kanban approach has been further developed in the past few years, no specific practices for its implementation exist, as illustrated by the different perceptions held among scholars about the approach itself. This has created the need for more study and analysis. The Kanban approach fundamentally depends on the lean principles that have been developed over the past 60 years; yet even these principles have no pure definitions in the literature available on the subject (Shah and Ward 2007; Al-Baik and Miller 2014). For instance, Womack and Jones (2003) mention only 5 principles, while Liker (2004) holds that there are 14 principles.

Wang et al. (2012) put credible efforts into identifying the lean concepts, principles, and practices, in which they, on one hand, identified five principles related to the Kanban approach based around Anderson's definition of the Kanban method. On the other hand, they identified a set of practices related to implementing these principles to software development. The application strategies of Kanban to software product development are defined as practices to implement the principle in specific settings. That is, a context-dependent implementation should differ from implementations in the manufacturing setting.

This study began as an attempt to shed some light on the practices involved in implementing the Kanban approach to software product development. The results of this study are expected to help establish knowledge of the different elements of the Kanban approach. The study also attempts to offer a first step towards developing guidelines for practitioners to use when implementing the Kanban approach in software development organizations.

Anderson's initiative has created a growing interest in the Kanban approach and how it can be implemented and set up in the IT industry. Even though kanban has been used for many years by Toyota in the TPS, the concept is relatively new in IT and still requires considerable study and investigation (Womack and Jones 2003). As a result, many IT organizations have been shying away from its use, while those who are using Kanban continue to struggle with the challenges the approach presents.

The scope of this review is specific to the implementation of the Kanban approach for product development processes in software development. In an attempt to develop a comprehensive Kanban approach, the investigators therefore define the Kanban approach as the following: *a set of concepts, principles, practices, techniques, and tools for managing the product development process with an emphasis on the continual delivery of value to customers, while promoting ongoing learning and continuous improvements.*

The scope and coverage of this systematic review differ significantly from previous reviews. For example, Dybå and Dingsøyr (2008) performed a systematic literature review of various empirical studies conducted on agile methods and lean software development up to the year 2005, and they were able to identify 36 relevant studies. Of those 36 studies, only a

single study reported on the application of lean practices to software development. They also found that these empirical studies focused on only one development method such as XP. Furthermore, the implementation of the lean approach was only carried out on a small scale with only three papers investigating organizations with more than 50 employees.

Another systematic review conducted by Wang et al. (2012) was limited to experience reports that had been published in agile-related conferences between 2000 and 2011. The scope of Wang's review was very wide and generic; it investigated the application strategies of more than 30 lean concepts, practices, and principles in agile software development. The review illustrated that organizations were mainly focusing on two lean concepts, "value" and "eliminating waste."

Wang et al. (2012) reported that organizations were moving from agile processes to lean processes, and this shift has led Wang to commend the development of operational guidance for each specific lean process, which was suggested as a significant area for future research and investigation. In short, Wang et al. (2012) do not specifically address the question of "What is Kanban?" in any detail. Table 1 emphasizes the differences between this systematic review given in this paper and the Wang et al. (2012) systematic review:

The results have been analyzed using the broad lean perspective, which relates to lean *product* development rather than lean *software* development. The systematic review approach of this study will also provide software development teams with an assessment of the methodological quality, and the strength of the evidence provided will deliver the necessary information prior to Kanban adoption.

3 Research Methodology

This research study has been undertaken by following the guidelines produced by Kitchenham and Charters (2007). The study has been conducted in three phases, 1) Planning the review, 2) Conducting the review, and 3) Reporting the review. Each phase has specific steps and outcomes, these steps and outcomes, specific to this research, are outlined in Table 2.

Table 1 Comparison with Wang et al. (2012)

Comparison element	The Kanban Systematic Review Al-Baik and Miller (2014)	Leagile Systematic Review Wang et al. (2012)
Purpose	1. Illuminating the guiding principles and elements of the Kanban approach to increase the likelihood of implementing it successfully by: 2. Defining the prime elements of the Kanban approach, and 3. Outlining the benefits and challenges of using the approach within IT organizations.	1. "Providing a better understanding of the application strategies of lean approaches in agile software development, and 2. Demonstrating how these strategies are being implemented in practice" Wang et al. (2012, p. 1288).
Years included	1990–2012	2000–2011
Scope	Specific to the Kanban approach	More than 30 different lean concepts, principles, and practices
Sources of primary studies	Journal articles, and <i>Grey Literature</i> , including conference proceedings and digital libraries Scopus, Scirus, Google Scholar, ACM Digital Library, IEEE Xplore, Science Direct (Elsevier), Web of Science, InterScience (Wiley), Science.gov, and InfoQ.com	Limited to experience reports that were published in agile-related conferences "XP Conference series, Agile Conference series, and XP/Agile Universe series" Wang et al. (2012)

Table 2 Steps and Artifacts of Systematic Literature Review

Phase	Steps	Artifact Outcomes
Planning Phase	- Identifying the objectives of the review	- Review Protocol document
	- Formulating the research questions	- Data Extraction Form
	- Developing and Evaluating the review protocol	
Conducting Phase	- Developing a search strategy and search strings	- Search Strategy document
	- Identify sources of <i>primary studies</i>	- Quality Instrument
	- Assessing quality of, and selecting, primary studies	- Data Extraction form (revised)
	- Extracting data from the selected primary studies	- Result summary
	- Synthesizing the extracted data	
Reporting Phase	- Presenting the results	- Systematic Review report
	- Writing and formatting the systematic review	
	- Evaluating the systematic review	

3.1 Objectives of Systematic Literature Review

The purpose of this paper is to illuminate the guiding principles and elements of the Kanban approach to increase the likelihood of implementing it successfully by defining the prime elements of Kanban and outlining the advantages and benefits of using it within IT organizations. The paper uses a systematic review methodology to look at the associated literature on the Kanban approach and its application in industrial I.T. settings. There is a need to empirically show the advantages that the Kanban approach can bring to IT organizations, which may increase confidence in the approach that in turn may lead to its wider adoption.

As recommended by Kitchenham and Charters (2007), the research group has framed the research questions based upon Petticrew and Roberts's (2006) criteria (*Population, Intervention, Comparison, Outcomes, and Context*) **PICOC**. The *population* for our study is the application area, of the Kanban approach, that is the IT settings for software product development; the *intervention* for this research is the implementation of the Kanban approach in software product development, while the main *comparison* is with agile methodologies in general and Scrum in particular.

The *outcome* of this study aims to illuminate the guiding principles to implement the Kanban approach in IT organizations to reduce both, cost and product's cycle-time, while increasing both, product quality and customer satisfaction. The *context* of this study targets industrial IT settings for software product development without a restriction on the size. The industrial settings might not be a full empirical study in one particular organization or project; for instance, Poppendieck and Poppendieck (2003) use several examples from diverse projects in different organizations throughout their book. The primary questions of this study are as follows:

- RQ.1. What elements of the Kanban approach are being discussed by the available literature as providing evidence of perceived benefits?
- RQ.2. What are the perceived benefits and challenges of using the Kanban approach, according to the available literature?
- RQ.3. Are the established guiding principles of the Kanban elements by the available literature well defined and applicable?
- RQ.4. Are agile techniques being relabeled as elements of the Kanban approach?

The first research question seeks to investigate Kanban's primary elements that the past studies have covered. It answers the following questions:

- RQ 1.1 What are the prime elements of the Kanban approach that have been discussed by the available literature?
- RQ 1.2 What are the relationships between the Kanban prime elements and the five core pillars of lean thinking?

Womack and Jones (2003) defined the five core pillars of lean thinking as:

- 1) *Value*: specifically the value from the customer's point of view, and value should be expressed in terms of a specific product or service.
- 2) *Value Stream*: identify the set of actions or steps required to transform the *value* from concept into finished product or service in the customer's hand.
- 3) *Flow*: once the *value stream* is well specified, and the wastefulness steps have been eliminated, make the *value* adding actions, or steps, *flow* without interruption or delay.
- 4) *Pull*: when *flow* is realized, a period time required to transform a concept into a finished *value* (product or service). Hence, the *value* has to be only introduced per the customer's demand. The *value stream* is triggered when the customer pulls the *value* when needed.
- 5) *Perfection*: when an organization implements the previous four principles, it realizes that there is no end for continuous improvement to seek *perfection*.

The second research question (RQ.2) seeks to investigate the perceived benefits and challenges that the available studies have reported. It answers the following questions:

- RQ 2.1 What are the perceived benefits of using the Kanban approach, according to the available literature?
- RQ 2.2 What are the perceived challenges of using the Kanban approach, according to the available literature?

The third research question (RQ.3) investigates the clarity and applicability of the established guiding principles as they have been undertaken by past studies to lead the Kanban initiatives in IT organizations. It answers the following question:

- RQ 3.1 Have the practitioners and scholars distinguished between the Kanban (capital K) method as a comprehensive change catalyst for software development and the kanban system (small k) as a pull system?

Finally, the fourth research question (RQ.4) assesses the relevance of results that have been reported in the past studies by examining whether the agile-related techniques were just relabeled as being the Kanban techniques. It considers the following question:

- RQ 4.1 Are Scrum boards being relabeled as Kanban boards?

3.2 Search for Primary Studies

In an attempt to perform a comprehensive scholarly and scientific search, and to minimize the bias into this systematic review, the search for *primary studies* was not only limited to

published journal article, but also targeted *Grey Literature*, including conference proceedings, digital libraries, and theses *as recommended by* York University, Centre for Reviews and Dissemination (CRD) Database of Abstract of Review of Effects (DARE).

The research has a planned systematic approach and specific sources to search for the *grey literature* to overcome the difficulties associated with finding articles to make it easier for the reader to verify the resources used in this research. The research contributions are limited from the year 1990 onwards, as Womack, Jones, and Roos (1990) introduced the concept of Lean in 1990. The search was only applied to abstracts, keywords and titles.

3.2.1 Sources of Primary Studies

As recommended by Kitchenham and Charters (2007), a combination of electronic search and manual search was conducted to minimize the threat of missing relevant studies. The Kanban approach and Lean have been used by IT organizations in different business domains, including but not limited to: IT Retailing, IT consulting, IT Quality Management, Software Development and Information Technology; thus, the related studies may be located in any IT related source or digital library; hence, the need has become apparent to search different electronic sources and digital libraries as there is no one authoritative source covering all the IT fields.

In an attempt to cover as many related sources as possible, the search covers the digital libraries and data sources as recommended by Brereton et al. (2007) in Kitchenham and Charters (2007). Table 3 provides a list of sources, which have been electronically searched to find *primary studies*, as well as the rationale behind searching each source.

Specific journals, such as “Empirical Software Engineering”, “The Journal of Systems and Software” and “The Information and Software Technology Journal”, were searched manually, as they are known to have either empirical studies or literature reviews and have been used by other *Software Engineering* systematic reviews. The full list of the manually searched journals and conference proceedings has been declared in the *research protocol*.

3.2.2 Search Strings

A list of terms was created earlier during the development of the *research protocol*; the list has been used to develop the search strings. The terms have been combined using the Boolean expressions (AND, OR) while being searched so as to reduce the number of irrelevant results that might not relate to the research area. That is, Google Scholar returned 32,200 results for searching (Kanban), 15,400 results for (Kanban AND Software) and 2,290 results for (Kanban AND “Software Development”). This is due to the fact that individually, the terms are also applicable to other disciplines.

The search strings were constructed based upon the structured research questions by following the *PICOC* method (see section 2.1), major synonyms for each element were combined using the *Boolean* expression *OR*, then the final research string was assembled using the *Boolean* expression *AND* as shown in Table 4.

The strategy for constructing the search strings was unified regardless of literature type, (*Monographic Literature*, *Grey Literature*, or *Systematic Literature Review*); however, the search terms were not the same. For instance, the *context* to search for primary studies in the *monographic literature* was composed of alternative terms to IT industrial settings or empirical evidence; however, the *context* to locate *systematic reviews* was composed of synonyms given to the systematic review as defined by Biolchini et al. (2005). The investigators have also reviewed the cited resources in the primary studies and reviewed these citations to include the studies that satisfy the inclusion criteria within the scope of the research area.

Table 3 Sources of Primary Studies

Source	Rationale	Literature Type
Scopus	Claims to be the world's largest database of abstracts and citations, with 49 million records including 5.3 conference papers	Monographic, Grey
Scirus	Claims to be the most comprehensive scientific search tool, with 545 million items including repositories and web information	Monographic, Grey
Google Scholar	Indexes the text of scholarly literature across many disciplines and sources. It covers theses, books, and publications from professional societies and online repositories	Monographic, Grey
ACM Digital Library	Claims to be the world's largest education and scientific computing society, it covers books, periodicals, reports and theses	Monographic
IEEE Xplore	A major resource for scientific and technical content. It claims that it expands its database by 25,000 new documents each month	Monographic
Science Direct (Elsevier)	Claims to contain more than 11 million full text articles of more than 2,500 journals and 11,000 books	Monographic
Web of Science	Provides access to cross disciplines and multiple databases. It provides a citation index to reference any other literature referencing the work of the searched study.	Monographic
InterScience (Wiley)	Claims to be the largest publisher for professional and scholarly societies. Provides access to over 4 million articles from 1,500 journals, 9,000 books, and reference works and databases.	Monographic
Science.gov	A gateway to over 2,100 scientific websites, provides search of over 55 scientific databases and 200 million pages of science information and research results	Grey
InfoQ	An online practitioner-driven community providing news, articles, presentations, and interviews to spread the knowledge and innovation in the software development community	Grey

3.3 Selection of Primary Studies

The selection of candidate primary studies was derived from the inclusion criteria that had been defined in the research protocol. Due to the enormous number of candidate studies and the limited time and resources available, the screening process consisted of three main steps. First, depending on the inclusion criteria as the solely determinant to whether to include or to exclude the paper, the authors screened the titles and keywords to determine the relevance to the study area. In case of uncertainties, the second step would be triggered, which involved screening the abstract of the paper, after which the investigators included or excluded the paper. If doubt still existed, step three would be carried out, reading the introduction and conclusion of the paper, then the investigators included or excluded the paper. The selection process that was followed is illustrated in Fig. 1.

3.3.1 Inclusion and Exclusion Criteria

For the purpose of this research study, *exclusion criteria* have not been established, the only *exclusion criterion* was “*If the study does not meet the inclusion criteria, it must be excluded*”, the following inclusion criteria have been established for this study:

Table 4 Search Strings

Element	Monographic	Systematic Literature
Population	“Software engineering” OR “Software development” OR “Information technology” OR “Computer science” OR “IT project management” OR “Software project management” OR “Information technology project management” OR “Software product development” OR “IT Product development”	“Software engineering” OR “Software development” OR “Information technology” OR “Computer science” OR “IT project management” OR “Software project management” OR “Information technology project management” OR “Software product development” OR “IT Product development”
Intervention	Kanban OR Lean OR “Continuous flow” OR “Pull system” OR “Lead time” OR “Cycle time” OR “Work in progress” OR “Work in process” OR WIP	Kanban OR Lean OR “Continuous flow” OR “Pull system” OR “Lead time” OR “Cycle time” OR “Work in progress” OR “Work in process” OR WIP
Comparison	Agile OR Agility OR Scrum OR Leagile OR Scrumban	Agile OR Agility OR Scrum OR Leagile OR Scrumban
Outcome	“Process improvement” OR “Cost reduction” OR “Reducing Cost” OR “Reduce cost” OR “High quality” OR “Higher quality” OR “Quality improvement” OR “Improving quality” OR “Improve quality”	“Process improvement” OR “Cost reduction” OR “Reducing Cost” OR “Reduce cost” OR “High quality” OR “Higher quality” OR “Quality improvement” OR “Improving quality” OR “Improve quality”
Context	Empirical OR Experiment OR “Evidence based” OR “Industrial setting” OR “Case study” OR “Action research”	“Literature Review” OR Overview OR “Research review” OR “Research synthesis” OR “Research integration” OR “Systematic review” OR “Integrative research review” OR “Integrative review”

- The study should be written in English.
- The study should be published between 1990 and 2012.
- The study should include empirical evidences derived from industrial settings, regardless of research methodology espoused in the study, i.e. *exploratory* or *explanatory*.
- The study should clearly state that it has its focus on software product development through the use of the Kanban approach or lean.
- The study should describe the elements and the approach used to implement Kanban.
- If the study has been published in more than one journal/conference, the most recent version of the study has to be included.

3.3.2 Study Quality Assessment

Each of the studies that had been selected previously was subjected to a full quality assessment process; the quality assessment process was developed to accommodate the three types of literature covered by this study; *monographic literature*, *systematic literature review*, and *grey literature*. As recommended by Kitchenham and Charters (2007), the research investigators developed a *quality instrument* to be the basis for the quality assessment (Appendix 2 describes the instrument).

Monographic quality was assessed by investigating the study’s rigor and relevance using the guidelines proposed by Ivarsson and Gorschek (2010), while *systematic literature review* was assessed as per the criteria proposed by York University, Centre for Reviews and Dissemination (CRD) Database of Abstract of Review of Effects (DARE). As there are no clear criteria to assess the quality of the *grey literature*, the investigators followed a systematic

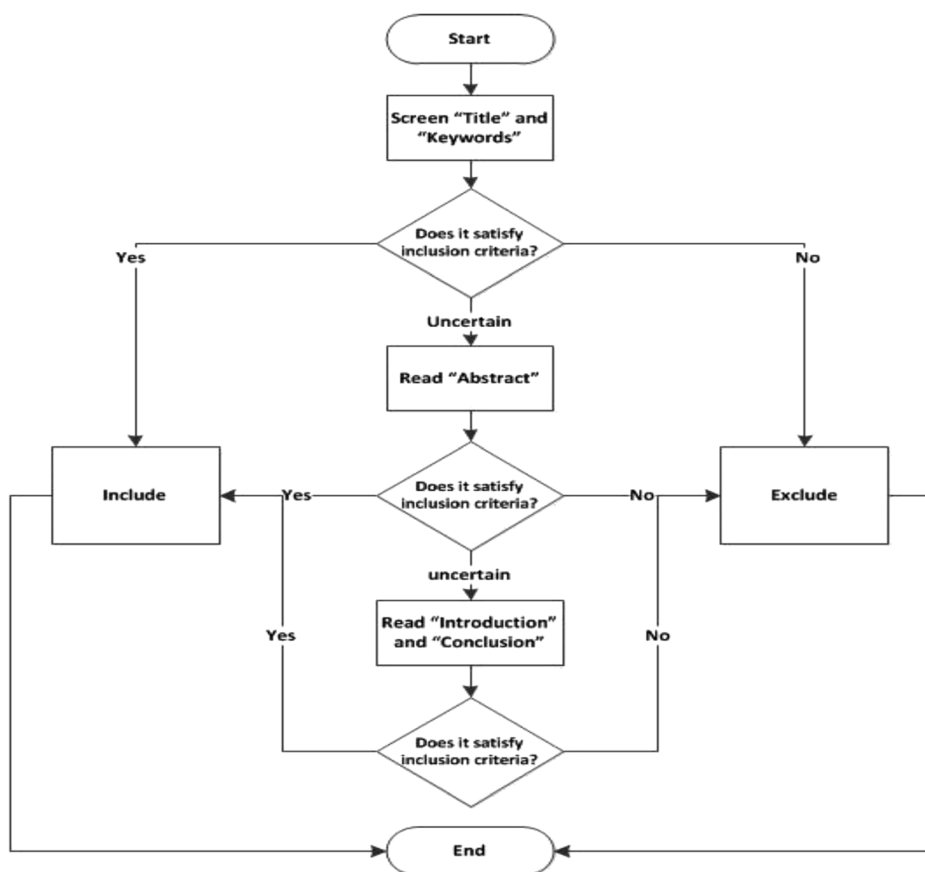


Fig. 1 Selection Process of Primary Studies

approach to assess the quality using two attributes: 1) the number of citation and 2) the *h-index* of the author(s). The studies that had not succeeded to meet a minimum level of quality threshold were excluded (see Appendix 3 for the list of excluded papers).

3.3.3 Reliability of Inclusion Decisions

Two research investigators have conducted the research study; hence, dispute and disagreement to include *primary studies* has occurred. The dispute between the investigators has been addressed scientifically by adhering to, and following, the *primary study* selection process presented in Fig. 1. Agreements between the investigators have been measured using Cohen's Kappa (*k*) method to estimate the *interrater reliability* (Banerjee, Capozzoli, McSweeney, and Sinha 1999). Scholars have not determined the level of *interrater reliability* that must be reached to consider the reliability measurement acceptable. Banerjee et al. (1999), however, suggested that for the interpretation of kappa values

For most purposes, values greater than 0.75 or so may be taken to represent excellent agreement beyond chance... and values between 0.40 and 0.75 may be taken to represent fair to good agreement (Banerjee et al. 1999, p. 604)

Hence, the research investigators decided to adopt the rigorous requirements of achieving greater than 0.75 on Kappa's measurements. The *inclusion criteria* were pretested twice, after first pretest, the *inclusion criteria* were revised and the pretest was conducted again on the same studies to assess the clarity of the *inclusion criteria*. The *interrater reliability* has been increased by 20 percent as a result of resolving one disagreement between the research investigators. After the second pretest, the criteria have been considered to be adequately reliable to proceed further. Table 5 shows the results of *interrater reliability* estimations.

3.4 Data Extraction

The data extraction stage aims to design the extraction form, which helps the research investigators to report the information that they will gain from the *primary studies* to answer the research questions. In an attempt to reduce the bias and as recommended by Kitchenham and Charters (2007), the extraction form has been defined and piloted during the *planning the review* phase as part of the *systematic review protocol*. The pilot process has not only helped in reducing the bias, but also, has given the investigators a chance to familiarize themselves with the data extraction process and to revise the form as required.

3.4.1 Data Extraction Procedures

The research investigators applied multiple data extraction methods to ensure the data extraction consistency. The methods were *test-retest* and *cross checking*, in which the primary investigator and the coauthor, respectively, performed a second extraction, and crosschecked twelve randomly selected *primary studies*. After applying the *inclusion criteria* on more than 3,000 studies, the candidate studies for inclusion had been reduced to 54 studies that went through the quality assessment process (section 3.3.2). As the investigators had to read the full-text of the published studies to determine its quality, the data extraction process was undertaken simultaneously, as the information would be still fresh in the investigators' minds. However, for the studies from *grey literature*, quality assessment and data extraction processes were totally separated as reading the full-text was not necessary to estimate the quality of the *primary study*.

3.5 Data Synthesis

In software engineering, data synthesis concerns with summarizing the results of the primary studies in order to help practitioners to adopt the appropriate technology (Biolchini et al. 2005). Two types of synthesis have been espoused by scholars in the field of Software Engineering, 1) *Descriptive* (qualitative) *Synthesis*, and 2) *Quantitative Synthesis*. Several methods for the synthesis are explained by Dixon-Woods et al. (2005), these methods are categorized into three main categories:

Table 5 Pretest Interrater Reliability

Pretest	Number of Studies Included	Interrater Reliability
Pretest 1–1	10	0.60
Pretest 1–2	10	0.80
Pretest 2	15	0.85

1. *Theory-led approaches*, which are intended to find relationships between findings from different studies;
2. *Analytical approaches*, which are intended to analyze the textual data reported by different studies; and
3. *Triangulation approaches*, which aim to draw more evidence by comparing qualitative and quantitative data from different studies.

For the purpose of this research and the type of data presented in the primary studies, the descriptive synthesis using the *analytical approaches* is considered the most appropriate type for the data synthesis. *Analytical approaches* include but not limited to, *grounded theory*, *meta-ethnography*, *Bayesian's meta-analysis*, *Miles and Huberman's data-analysis*, and *content survey*. Given the research questions of this study, the latter method is determined to be the most appropriate method to synthesis the data to answer the research questions.

3.5.1 Case Survey Approach

We employed the extended version of the *case survey* proposed by Jensen and Rodgers (2002) for categorizing and coding related data from qualitative studies for quantitative analysis. The data is summarized in a *cross-case* summary table, in which each row represents a *primary study*, and each column represents a specific element related to the Kanban approach, the cell entry is a color-code indicating that whether a particular study supported the given variable. The *case survey* method is a very powerful method to synthesis both qualitative and quantitative evidence (Dixon-Woods et al. 2005).

3.5.2 Coding Mechanism

The *coding scheme* for categorizing the components of Kanban was determined by using *reciprocal translation* suggested by Kitchenham and Charters (2007), specifically the *Meta-ethnography* (Dixon-Woods et al. 2005). The investigators examined the recurring themes in the articles identified as primary studies in the current literature to draft the preliminarily *coding scheme*. Each category was well defined and communicated to the research panel (peer-reviewers familiar with the research area) that provided feedback, and then the *coding scheme* was revised (see Appendix 4 for description of the coding scheme).

Likewise the inclusion criteria, the *coding scheme* was pretested twice, following the first pretest, the *coding scheme* was revised and the second pretest was conducted on the same studies to validate the *coding scheme* clarity and applicability. The co-author applied the *crosschecking* technique and coded fifteen of the primary studies independently using the coding scheme. The *interrater reliability* of the coding is shown below in Table 6.

Table 6 Coding Interrater Reliability

Pretest	Number of Studies Included	Interrater Reliability
Pretest 1	10	0.73
Pretest 2	10	0.86
Cross-Check	15	0.92

4 Results

This section is a presentation of the results of the systematic literature review derived from 37 papers that were finally selected as per the research questions specified earlier. In addition to the findings related to the Kanban approach, this section provides an insight into the results of the search process as well as the quality assessment of the *primary studies*.

4.1 Results of Search and Quality Assessment

The search generated 7,809 articles that were written between 1990 and 2012. Of those, 4,568 articles were identified as duplications through the use of the citation and bibliography management software package, Mendeley, which reduced the final number to 3,242 articles for screening.

After applying the inclusion criteria, 3,117 articles were found to be irrelevant or written in languages other than English, which limited the number of candidate articles to 126 articles. Finally, 72 candidate studies were excluded because there was a more recent or complete copy, the full text of the article was not available, or the focus of the article was not on software or Kanban. As a result, a total of 54 articles were left for full-text reading, quality assessment, and data extraction.

After analysing, estimating, and assessing the quality of the candidate studies, 17 were excluded for either being irrelevant or not meeting the minimum quality threshold (see Appendix 3 for the list of excluded studies). Hence, the primary study list has 37 studies (S1–S37; see Appendix 1 or the list of included studies). The distribution of the primary studies per study type (i.e., Journal, Conference, Web Article, Thesis, or Book) is illustrated in Fig. 2.

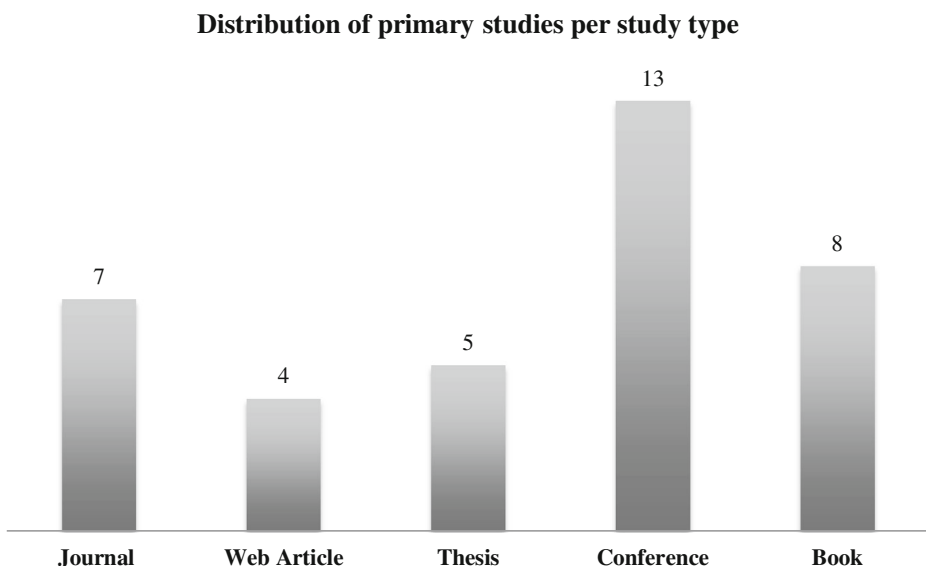


Fig. 2 Distribution of primary studies per study type

4.2 Research Results

Given the variety of primary study types and research contexts, the primary studies have been divided into four main categories based upon similarities of the study type and context. Table 7 shows the four categories; this categorization has helped in justifying the scholars' position towards each element of the Kanban approach within each category. For example, Type A scholars see *planning* as a source of waste as it does not add value from the customer's point of view; Type B scholars, however, advocate the use of *retrospective planning* to facilitate the achievement of continuous flow.

Figure 3 below shows the distribution of primary studies per category per type. The distribution revealed some important conclusions based upon the type of the study conducted. For example, theses in general seek to prove specific phenomena. In this particular study, theses seek to prove that Kanban extends beyond agility as a stand-alone methodology, which indicates that Kanban is not yet a well-defined methodology. On the other hand, as shown in Fig. 3, conferences tend towards Type C, which means that agile practitioners and scholars are looking into enhancing the existing agile methodologies by implementing the Kanban approach.

Distribution of the primary studies per category per year is illustrated in Table 8. It shows that the Kanban approach is relatively new in IT and still needs considerable study and investigation. The focus on tailoring the Kanban approach to software development was started in 2003, however, it has increased since 2009 with no clear direction regarding the elements of the Kanban approach as it offers a less prescriptive element as compared to agile.

4.2.1 Kanban Elements

The analysis of the 37 studies (S1–S37) provided 20 different concepts, principles, and techniques related to Kanban that have been identified as the Kanban elements. This wide set of elements resulted from the different perspectives based upon the categories explained above. In Figs. 4, 5, 6 and 7, the headers (C1–C20) represent the 20 elements of the Kanban approach, which are described below.

A three-color scale was used to visually represent the 20 elements. The green colour represents concepts that were discussed and thoroughly described; as such, we refer to them as “Discussed”. The amber represents those concepts that were merely mentioned; for

Table 7 Categories of Applying Kanban to Software Organizations

Type	Category Name	Category Description
A	Kanban as an element of lean (beyond agile)	This category considers Kanban one of the primary elements of lean. Kanban here is used as a starting point or a first step towards transforming the organization from agile to lean.
B	Kanban as an agile methodology	This category considers Kanban as an agile methodology. Kanban is used here to realize the agile manifesto.
C	Hybrid of Kanban and Scrum	This category is considered a combination of both Kanban and Scrum. Kanban has been used to overcome the limitations and leverage the advantages of Scrum.
D	Explanatory Kanban	This category explains Kanban and its related concepts and elements as a stand-alone tool without direct linkage to either agile or lean.

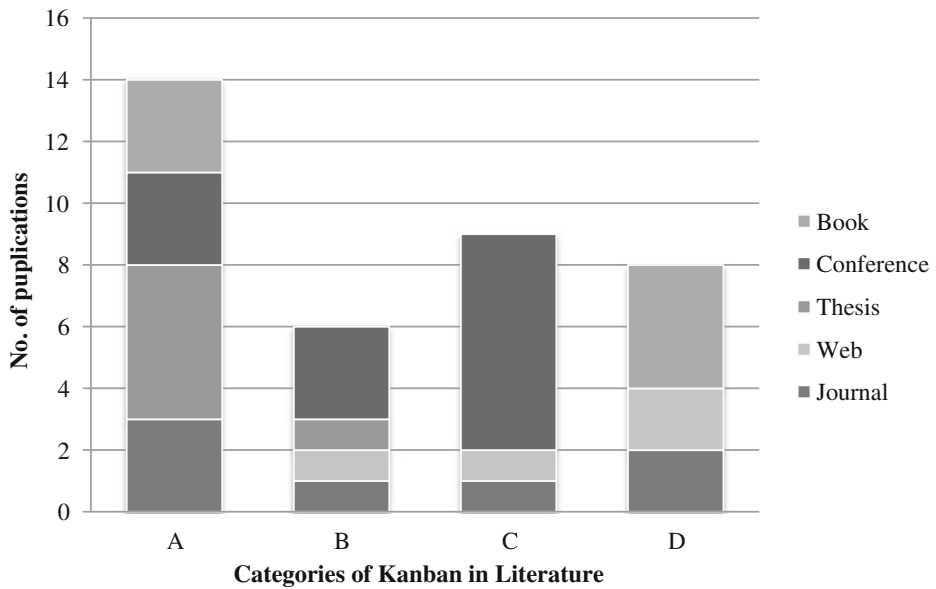


Fig. 3 Distribution of primary studies per category per type

consistency, we refer to them as “Stated.” The red represents the concepts that were not discussed at all; therefore, we refer to them as “Not Mentioned.”

From Figs. 4, 5, 6 and 7, it is evident that some Kanban elements were established and discussed more thoroughly than other elements regardless of the study context and type. Scholars in a specific category discussed some elements that were not mentioned at all or were touched lightly by scholars in other categories. A summary of the 20 elements is outlined below:

C1 *Kanban method*, describes the methodology to implement Kanban in an organization. Kanban should be implemented in an incremental and evolutionary fashion, the method should also be reviewed and continuously improved [S2, S4–S8, S11, S12, S19, S21, S29, S31, S33], which is referred to as “*Kaizen*” by lean practitioners.

Table 8 Distribution of the primary studies per category per year

Year	Type A	Type B	Type C	Type D
2003	[S5]			[S23]
2007		[S2]		[S36]
2008		[S28]		
2009	[S6] [S31]		[S11]	[S35]
2010	[S14][S18][S25]		[S15]	[S19]
2011	[S7][S21][S24] [S26][S43]	[S3] [S9] [S32]	[S4] [S17] [S32]	
2012	[S22][S27][S29]	[S30]	[S10] [S8] [S12] [S36]	[S1] [S13] [S16] [S20]

ID	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18	C19	C20
S5	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S6	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S7	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S14	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S18	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S21	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S22	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S24	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S25	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S26	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S27	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S29	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S31	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S34	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●

Fig. 4 Cross-Case summary of the Kanban elements for Type A

C2 Kanban board, the tool to visualize the workflow. It represents the activities to be performed from product/service conceptualization until it reaches the customer's hand. Some scholars recommended a design board based upon the existing activities before improving the process evolutionarily [S2, S5, S7], whilst other scholars [S3, S6, S22, S29] used the traditional waterfall approach as the basis for the board design. Still other scholars [S12, S21, 34] used a simpler form of board design to include three main queues (i.e., To Do, In Progress, and Done).

C3 Pull system, the approach of starting the development process based upon a request from a customer [S5, S6]. It also requires *work-in-progress* (WIP) limit to be set out, as a work item should be pulled only if the WIP limit is not reached [S2, 29]. In addition, it is recommended to pull the highest priority items, if possible [S24].

C4 Prioritized queue, which refers to the list of ordered requirements or work items. Several priority criteria were set by different scholars, [S29] suggested that organizations prioritize work items based upon their value, urgency, and importance. [S2] set priority based upon *cost of delay* or resources availability. [S26, S4] defined several queues for each process in the flow by placing a queue for each process in the Kanban board.

C5 Inclusion criteria, one of the major elements of the Kanban approach. These criteria guarantee that the work will be of value for the customer before being added to the board. None of the studies has attempted to define the inclusion criteria, except [S4] which shed some light on how they prioritize the work items and what were the criteria to add the work items to the waiting queue.

C6 Work-in-progress (WIP), the concurrent number of work items that is allowed for each process in the workflow. WIP has been identified as a major characteristic of the Kanban approach and the main objective of Kanban (Wang et al. 2012). The concept, however, has not been clearly defined. For instance, majority of the studies suggested that organizations set WIP

ID	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18	C19	C20
S2	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S3	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S9	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S28	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S30	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S32	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●

Fig. 5 Cross-Case summary of the Kanban elements for Type B

ID	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18	C19	C20
S4	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S8	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S10	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S11	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S12	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S15	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S17	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S33	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S37	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●

Fig. 6 Cross-Case summary of the Kanban elements for Type C

limit by experiment [S1–S5, S7, S9, S10, S12–S14, S16, S18–S24, S26, S27, S29–S34, S37], while some studies [S11, S25, S36] did not mention WIP at all.

C7 Done Item, refers to when a work item is considered completed. This element should have been investigated more systematically as it is considered one of the major contributors to the continuous flow. Ironically, the majority of the existing primary studies [S1, S4–S7, S11, S13, S14, S16, S19, S22, S24–S26, S28–S33, S35–S37] did not mention this concept. [S9, S12, S27] claim that they have established a *done item*; however, none of the studies offered further explanation of what that definition was or how it was established. In [S17], the *done item* was established as checklists based upon code quality guidelines and standards.

C8 Reverse item, which is a work item that is being moved backward on the Kanban board instead of moving forward, for instance, a bug fixing work item, which needs further development. This element has not been considered a major issue as the scholars who shed some light on this element suggested that the work item flows in loops [S5, S35]; Lundgren (as cited in [S3]) suggested putting the item back to the backlog. Hence, there is no such backward movement.

C9 Validated learning, which is a validation process for a feature that has been completed to measure its value from a business perspective (Ries 2011). The scholars in [S26] suggested that monitoring and learning is one of the key performance indicators to measure the project success.

C10 Cycle time/Lead time. This element is arguably the determinant for the process efficiency and effectiveness as it is used in measuring the performance of Kanban. Nevertheless, Scholars have not reached an agreement on its definition [S30]. Scholars used *cycle time* and *lead time* interchangeably [S14]. Some scholars [S5, S19, S20, S30] defined *cycle time* as the time elapsed from work start to end to produce a feature; however, some other scholars defined it as the time between deliveries of items [S23, S35].

C11 Performance measurement tool. This element describes how the performance of Kanban is measured. Some studies [S2, S5, S14] indicated the use of a *cumulative flow* diagram based

ID	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18	C19	C20
S1	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S13	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S16	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S19	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S20	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S23	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S35	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
S36	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●

Fig. 7 Cross-Case summary of the Kanban elements for Type D

upon cycle time and WIP. Other studies [S21, S25, S29] encouraged tracking the progress on a daily basis using *burn-down* charts.

C12 *Bottleneck*, a point of congestion where a processing point cannot keep pulling work items at the same pace of production from the preceding processing point. Bottlenecks can be identified by visualizing the workflow [S14, S21] or by establishing a measure to detect bottlenecks [S24] that was adopted from the *theory of constraints*. Zhang in [S18] suggested decomposing the bottlenecks into smaller chunks and realizing their expected value in increments.

C13 *Slack or Buffer*. The buffer is needed to determine the *waiting time* [S14] a work item stays in the queue before being processed, to handle a process's cycle-time variation [S29, S31], and to reduce the creation of bottlenecks. An appropriate *buffer* size should be introduced in front of the bottleneck process; the buffer size should be reviewed periodically and adjusted as needed [S29].

C14 *Waste identification*. In summary, waste is anything that does not contribute to the value creation for the customer; hence, it should be removed (Al-Baik and Miller 2014). Consensus has been reached on this element; several scholars [S2, S14, S19, S23, S24] agreed that Kanban helped in visual communication and identification of wasteful activities, so that wasted energy and productivity were reduced and more energy was put towards a quick response to customer requests.

C15 *Team collaboration*. Many software development teams lack collaboration, cohesion, and cohesiveness. IT teams are still perceived to operate as islands, and bridges between technical people and business people are needed [S29]. Oostvogel in [S16] argued that the minute a team is involved in a Kanban flow, their cooperation mode begins to change. Kanban increases communications between team members themselves on one side, and team members and senior management on the other side.

C16 *Meeting structure*. This element is one of the most arguable elements. On one hand, some scholars [S30] identified meetings as sources of waste that should be eliminated. On the other hand, other scholars [S2, S14, S24, S27, S31] recommended a daily stand-up meeting for 15 min to discuss constraints, while Kniberg in [S21] argued that each team should be given the freedom to decide meeting length and structure.

C17 *Avatars*, the visual representation of the task owner who is responsible for getting the work done. This visual demonstration makes Kanban a powerful tool in making informed decisions. The data presentation on the Kanban board makes it easier to make a factual-based decision [S14, S29]. Management can get information on resource capacity and availability that helps in resource assignment and scheduling by just looking at the Kanban board.

C18 *Planning and estimation*. This element is similar to C16, as there is a contradiction of views about its importance. Several scholars have identified planning as a source of waste, and have suggested that reducing this waste will increase both, productivity and customer satisfaction. For instance, the authors of [S30] claimed that by reducing the planning activities in their sprint-planning, the *lead-time* was reduced significantly. Zhang in [S18] reported that *planning* took two-thirds of the production process *lead-time*. However, Zhang in [S18] as well justified that planning and retrospective activities are important, as they provide an opportunity for improvement through continuous feedback from team members.

C19 *Policies*, which govern the Kanban approach, they are the dominant guidelines that outline what work should be done and how it should be done. Anderson [S5] claimed that *policies* help in producing a healthier environment to solve disputes and reach consensus. He also claimed that policies help in moving a discussion of problems from an emotional and subjective discussion to a rational and objective discussion. Boeg in [S29] suggested having the team members involved in developing the Kanban policies.

C20 Feedback loop. Getting feedback from customers is the primary goal from delivering a product through short cycles and frequent builds. Several scholars [S22, S26, S35] underlined the importance of a *feedback loop* in providing facts rather than guessing about how the customers perceive the product; however, no sufficient effort was put towards defining how Kanban would help in getting customers involved to provide their feedback.

4.2.2 Benefits and Challenges of Using the Kanban Approach

The primary studies reported various benefits and challenges associated with the use of the Kanban approach, and the investigators have summarized the benefits and challenges that were reported by at least five primary studies. Most of the studies reported benefits and challenges related to the management of the software development process as well as the software development process itself. Few studies covered the human aspect of the software development process. Other areas of software development reported by fewer than five studies included the configuration management, design, and maintenance.

Table 9 summarizes the benefits of using the Kanban approach to software product development. The most frequently reported benefits, which were reported by about 46 percent of the primary studies, were both, enhancing visual control to facilitate the management

Table 9 Benefits of the Kanban Approach

Benefits	Reporting Studies	Percentage of Reported Benefit
Enhancing visual control that facilitated and supported the decision-making process	[S2][S4][S10][S11][S13][S14][S15][S20][S21][S23][S24][S26][S29][S31][S35][S36][S37]	45.9 %
Facilitating the coordination of cross-functional teamwork and imposing self-organization	[S1][S2][S4][S5][S10][S14][S15][S16][S17][S19][S21][S22][S33][S34]	37.8 %
Empirically introducing quality circles and kaizen events	[S2][S5][S11][S14][S17][S19][S24][S25][S29][S34][S35]	29.7 %
Reducing the cycle time/lead time	[S2][S4][S15][S17][S18][S22][S24][S25][S29][S32][S37]	29.7 %
Increasing customer satisfaction and realizing high value	[S1][S11][S14][S15][S23][S24][S25][S26][S29][S35]	27.0 %
Decreasing market and technical risks of the product	[S2][S11][S16][S18][S24][S26][S30][S34][S35]	24.3 %
Developing continuous improvements strategies	[S1][S5][S7][S12][S13][S14][S15][S17][S19][21][S22][S23][S24][S27][S29][S33][S37]	45.9 %
Increasing the predictability in the delivery of the final products with the constraint of changing customer requirements	[S2][S4][S11][S14][S15][S23][S24][S26][S27][S30][S32][S33][S35]	35.1 %
Ensuring skills development and cohesiveness for teams	[S4][S5][S11][S15][S21][S33]	16.2 %
Driving and facilitating organizational change management and culture changes	[S2][S4][S5][S8][S13][S14][S19][S20][S21][S26][S33][S34]	32.4 %
Enhancing quality of product, indicated by decreasing the defects rate, increasing the quality assurance pass rate, and reducing the number of bugs.	[S11][S14][S24][S25][S30][S34]	16.2 %

decisions and developing strategies for continuous improvements. While the least-reported benefit, with 16.2 percent, was related to the team development and cohesiveness.

According to the available literature, Kanban implementation and use can greatly improve the development process; however, many teams are faced with challenges when using the approach. There issues are related to the team working together to successfully deliver a software product. The team should also understand the Kanban board and its different elements to ensure that they use them effectively to improve their software development process.

Table 10 emphasizes the challenges that accompanied the decision to use the Kanban approach. Of the primary studies, 51.4 percent perceived a challenge with the absence of guidelines for using the Kanban approach within IT settings. Only 13.5 percent reported the use of the Kanban approach in combination with agile techniques as one of the main challenges.

The conspicuous challenge reported by the studies is related to a lack of details and the absence of the guidelines on how the Kanban approach can be used in IT organizations, as well as the lack of guiding principles on how to introduce the different Kanban elements. The Kanban approach fundamentally depends on the lean principles that have been developed over the past 60 years; yet the available literature provides no pure definitions for these principles (Shah and Ward 2007; Al-Baik and Miller 2014).

4.2.3 Clarity of Available Guiding Principles

As stated previously, in an attempt to establish guidelines for the Kanban approach, Anderson has differentiated the *Kanban method* from kanban—the pull system—by capitalizing the word Kanban. Anderson distinguished between the Kanban method and the kanban pull system as follows:

Table 10 Challenges with the Kanban Approach

Challenges	Reporting Studies	Percentage of Reported Challenge
Defining the measurement metric, for instance, lead time has been deemed problematic to assess the performance, is difficult.	[S2][S5][S6][S16][S19][S24][S29][S30][S35][S36]	27.0 %
Implementing the Kanban approach requires deeper understanding of lean concepts, principles, and practices.	[S3][S7][S10][S18][S22][S23][S25][S28]	21.6 %
Unclear definitions of the Kanban prime elements are used, such as the definition of “Done.”	[S7][S8][S10][S12][S15][S17][S18][S23][S32][S37]	27.0 %
Guidelines to implement the Kanban approach and guiding principles on introducing the Kanban elements, such as WIP limits, are absent.	[S3][S4][S5][S6][S7][S8][S11][S12][S13][S15][S16][S17][S18][S23][S25][S27][S29][S32][S35]	51.4 %
IT organizations have difficulties in digesting some elements of the Kanban approach, such as flow.	[S3][S7][S8][S10][S12][S14][S16][S24][S25][S35]	27.0 %
Development teams need to be reorganized and restructured.	[S3][S4][S7][S10][S27][S33]	16.2 %
Priming the Kanban approach to software development process and integrating it with existing agile techniques is complicated, expensive, and time-consuming.	[S6][S9][S10][S12][S18]	13.5 %

Kanban (capital K) as the evolutionary change method that utilizes a kanban (small k) pull system, visualization, and other tools to catalyze the introduction of the Lean ideas into technology development and IT operations. [S5, p. 6]

Anderson clearly stated that Kanban is neither a software development life cycle nor a project management methodology; instead, Kanban is used to incrementally improve an existing process. Kanban suggests customizing the process template to suit the organization's needs; it does not recommend adopting a defined method or template. Due to this distinction, Kanban has become a controversial in the agile software community (Anderson 2010).

The analysis of the primary studies revealed the reasons behind the dilemma that practitioners and scholars have been experiencing with the established guiding principles in the domain. Although several studies have conformed to Anderson's definitions of small "k" kanban and capital "K" Kanban as reported in [S6] and [S18] in which they distinguished between Kanban as a change method and kanban as a pull system, Kanban was defined as a learning tool to incrementally introduce lean principles (pull, flow, and value). [S4] reported the use of Kanban to modify Scrum by eliminating the iterative development and burn down chart and introducing several prioritization queues.

Several studies ([S1][S7][S8][S14][S22][S24]) have theoretically defined kanban (small k) as the pull system to realize the continuous production flow; however, [S1][S8][S22], in fact, have reported the practices of using Kanban (capital K) as a change method. [S8], for example, introduced Kanban to amend Scrum by eliminating sprint planning and time boxing, which are more related to the definition of the Kanban method not the pull system.

Another example was derived from [S1], where in the background section of the study, kanban (small k) was defined as a pull system that signals an upstream process to start production when a downstream process is ready to accept more work. Conversely, the reported findings were more related to the Kanban method (capital K). The authors of [S1] deemed the changes imposed by Kanban on the Scrum practices as an explicit rejection of Scrum. They also considered Kanban as a stand-alone lean methodology, which contradicts Anderson's statement that Kanban is not a software development life cycle.

Due to the nature of the work in HR, feasibility of implementing Scrum was rejected, including the concepts of time boxing, scheduled releases, story point estimation, sprint burn down, etc. Instead, kanban was seen a better fit for the HR department. [S1, p. 1292]

Several studies ([S3][S9][S12][15][S17][S18][S26][S27]) have agreed on three prescription rules for Kanban: 1) visualize the workflow, 2) limit WIP, and 3) measure the flow. In addition to these prescribed rules, [S21] and [S29] have added two guiding principles as predecessors for the prescriptions: 1) start by doing exactly what you are doing right now, and 2) map the value stream. This tends more towards the Kanban (capital K) method. However, [S3] and [S9] have described Kanban as an agile methodology and not as a method for change and improvement, which contradicts Anderson's description of the Kanban method.

In [S9], the authors reported what they claimed was a transition from Scrum to Kanban, "Process transition was made overnight, however, the Process Transition stage lasted for two weeks" [p. 160]. They claimed that the goal was to improve the software development process, but they reported a big bang transition from Scrum to Kanban, which violates the Anderson's Kanban rule of incremental change.

The authors in [S30] also reported a transition from Scrum to Kanban, "in 2010, the company switched from Scrum to Kanban" [S30, p. 48]. They reported amending some changes to Scrum by introducing Kanban: "Reduced sprint-planning activities (and abandoned cross-functional teams) by the end of 2009. Two employees mentioned this relaxation of

the Scrum rules as an explanation for why the lead time reduced from 2009 to 2010” [S30, p. 52]. However, they insisted on calling it a transition, “*after replacing Scrum with Kanban, SI almost halved its lead time, reduced the number of weighted bugs by 10%, and improved productivity*” [S30, p. 52].

The use of small “k” and capital “K” in the manuscript of the primary studies is another point worth mentioning. In addition to Anderson, only two studies ([S16] and [S34]) distinguished between the Kanban method and the kanban pull system. Two other studies ([S15] and [S37]) interchangeably used small “k” and capital “K.” Surprisingly, [S15] was forwarded by Anderson himself.

More than 40 percent of the primary studies ([S3][S4][S8][S9][S12][S14][S17][S22][S24][S27][S28][S30][S32][S33][S35]) did not use small “k” in their manuscripts at all. Other studies ([S6][S23][S25][S31]) followed the formal English writing by using capital “K” at the beginning of a new sentence and small “k” in the text. The authors of [S26] used small “k” and capital “K” in their own specific way: the former referred to kanban in manufacturing while the latter referred to Kanban and its elements in software development.

4.2.4 Kanban Board

The Kanban board, a tool in the Kanban approach, is used to visualize and coordinate the work of the software development teams. The tool features columns that illustrate a flow of activities and sticky notes to represent work items. For every activity in the process, limits are placed on the number of work items that will obtain an overall limited WIP (Ladas 2008).

The Kanban board can be used to enhance communication and flow of information within an organization. Employees can pin their feedback to the board while management can be able to respond and make changes within the organization (Pink 2009). In [S29], Boeg argued that the Kanban board is also used to illustrate issues, challenges, and bottlenecks. However, how the Kanban board can be used to help teams identifying these challenges and bottlenecks is poor illustrated.

The analysis of the primary studies showed that the Kanban board is not different from a Scrum board, also called a Scrum task board. At times, the primary studies explicitly stated this finding. For example, in [S15], Kniberg and Skarin imposed the question, “*What’s the difference between a Scrum board and a Kanban board?*” [S15, p. 15]. They then answered the question with, “*the little 2 in the middle column on the kanban board. That’s all.*” The number refers to the WIP limit.

A similar answer was provided by [S3]:

In Scrum, tasks are placed on boards for time-boxed sprints. Kanban is a different way to approach this. Instead of being time-boxed, tasks are pulled at any time, only limited by a work-in-progress (WIP) limit that restricts the allowed number of tasks in every workflow state. [p. 15]

The authors of [S12] reported that the Kanban board emerged from the changes that they had made to the Scrum board; however, the authors did not state the nature of these changes. Poppendieck and Cusumano in [S22] suggested:

“Kanban board might start out in a software development environment, but can easily expand to include more steps in the value stream, such as marketing and operations.” [p. 31]

However, this has not progressed beyond being a suggestion. Explaining what the Kanban board is and what it involves is not enough; more research and analysis must be carried out.

Although there are no specific rules regarding how a Kanban board should look, although there is a proposed Kanban board layout. The Kanban board has columns that show the activities as they are organized. The columns will then hold different activities and illustrations. It is necessary to clearly illustrate how the different Kanban board components work alongside each other and to describe the role of the team members in ensuring that this is occurs.

4.2.5 Interrelation between Kanban Elements and Lean Pillars

None of the scholars discussed the interrelations between the Kanban elements and either the lean pillars or the agile manifesto. As such, there is a gap in information on how the concepts interrelate as the Kanban approach is implemented in an organization. In an attempt to reduce this gap, the concepts are discussed together to give practitioners a better understanding of the relationships between the lean five pillars and the elements of the Kanban approach.

The associations were identified based upon the results of the *Thematic Analysis* that was espoused to specify the coding scheme. Two types of associations were identified; positive association and negative association, the latter was determined if at least one study reported a negative impact or disadvantage resulted from the implementation of that specific element of Kanaban. Figure 8 below depicts these relationships.

Based upon the findings that were reported by the included primary studies, the figure above shows that four out of five lean pillars (i.e., Value, Values Stream, Flow, and Pull) may be impacted negatively by implementing any of the following elements: C8, C13, C16, or C18. Care must be taken when recognizing the necessity of implementing these elements.

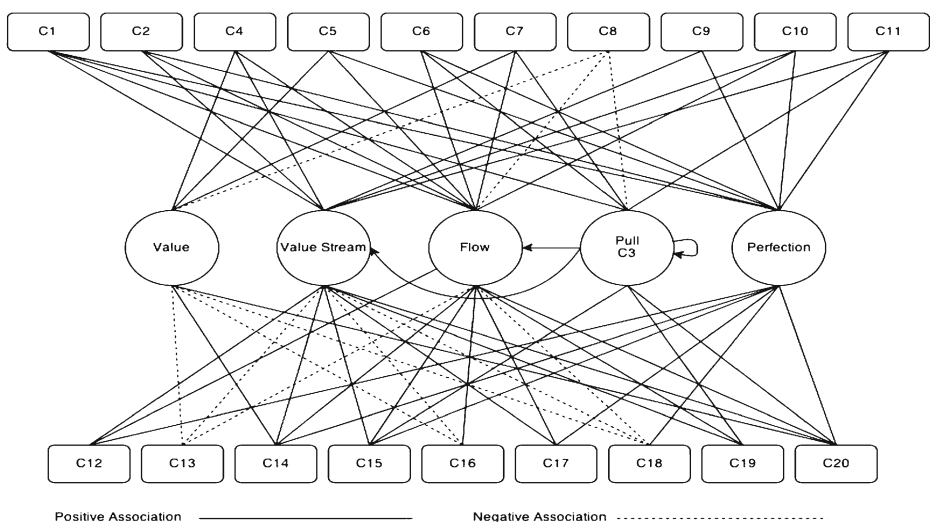


Fig. 8 Associations of Kanban concepts and Lean five pillars

5 Discussion

Kanban is heavily influenced by Anderson (2003) work with lean product development. Researchers have identified that it is a good fit to be used in the software development just as it has been used in maintenance and operation (Hibbs, Jewett, and Sullivan 2009). There is a misconception that Kanban and Scrum are similar with minor differences. In actuality, Kanban and Scrum are individual concepts as Kanban acts as a change agent and the principles in Scrum can be used to optimize the Kanban workflow (Poppendieck and Poppendieck 2003).

IT practitioners have started to use Kanban in the software development process as a means to optimize the workflow, whereby they aim to reduce wastes and to add value to the product development (Anderson 2010). However, the analysis of the primary studies revealed major issues in the existing literature that limit practitioners from implementing Kanban appropriately. The issues can be classified into three main taxonomies: 1) *definition disagreement*, which includes the elements that lack standard definition in the available literature, 2) *vague guidelines*, which consists of the elements that have poor instructions and guidelines on how to be designed and implemented, and 3) *contradiction and conflict*, which classifies a element that contradicts and conflicts with another element or with one of the lean five pillars.

The classification of issues to these taxonomies was not a simple and straightforward task to accomplish as one issue could be categorized under more than one taxonomy. For instance, C10 (cycle-time/lead-time), the analysis of the available literature has shown that scholars have not reached an agreement on a standard definition for any of the terminologies, which compromises a *definition disagreement*, consequently, there were no agreed upon instructions on how to design and measure either cycle-time or lead-time, which can be categorized as *vague guidelines*.

Likewise, *contradiction and conflict* issue was derived from the definition of *cycle-time* in [S30] in which the *cycle-time* was defined as the time elapsed from work start to end to produce a feature; the authors argued that *cycle-time* should not start from the minute a customer asks for a new feature because of two reasons, the first reason that a feature would have to wait for a while before start working on that feature, hence, waiting time in the backlog should not be counted towards the *cycle-time*; this reason contradicts with *waste identification* and violates the continuous *flow* pillar of lean. The second reason that any person other than the customer might propose a feature to be developed, which conflicts with pull, a prime element of Kanban and another pillar of lean.

One principal discussion point is the Kanban board design. The analysis of the included studies showed a theme of designing the board based upon the classical waterfall approach, which was clearly noted by Lundgren (as cited in [S3]). It is our believe that the recommendations of scholars in [S2, S5, S7] would be more applicable by representing the current activities and improve evolutionary, which is very much aligned with lean core pillars, *value stream* and *perfection*.

Another interesting argument is related to normalizing the work items in terms of size. Several scholars [S10, S15, S33] reported the use of cadence and team velocity to determine the amount of work that could be accomplished during a specific period of time, which implies the standardization of work items size. Henrik in [S15] clearly reports the decomposition of work into pieces of roughly the same size based upon the Minimum Marketable Feature (MMF). On the other hand, Reinertsen in [S35] argues that spending time and efforts to reduce variances of tasks instead of handling and managing these variances, is actually a wastefulness activity rather than a value adding activity to the product development lifecycle.

5.1 Proposed Guidelines to the Kanban Approach

Upon reviewing the findings, it was evident that the team members contribute a great deal to the Kanban implementation in an organization. The Kanban approach has been used by Toyota

in its TPS for many years now through the JIT approach. Consequently, due to its success in the TPS, many other manufacturing industries use the approach as a means to optimize the workflow. IT practitioners have started to use Kanban in software development to reduce waste and to add value to product development (Anderson 2010); however, the approach has not been fully understood. More than 30 percent of the primary studies identified three core principles, which are discussed below in greater details.

5.1.1 Visualize the Workflow

The recommendation of this paper is to start implementing the Kanban approach by visualizing the workflow; It's important to start by understanding how the current system works. One of the bases of the Kanban approach is the necessary emphasis on the current system because Kanban serves as a change agent not as a replacement for the whole process. Several studies ([S5][S7][S21][S29]) recommended visualizing the exiting development process and its activities, and then modifying the board as the process is improved and wastefulness activities are eliminated.

There are different ways to visualize the workflow to improve it; one of the approaches as recommended by more than 24 percent of the primary studies ([S1][S3][S7][S14][S15][S20][S22][S23][S24]) is through value stream mapping. Value stream mapping is the process through which the team identifies all the steps in the software development process in order to make a product that is desirable to the client. The term “stream” here refers to the fact that there is a smooth unbroken flow between varieties of steps. A continuous smooth flow of valuable new features and elements will lead to a desired final product.

For a value stream to lead to successful software product development, it is crucial for everyone to be considered, from the customers to the developers and support engineers not merely the development team (Womack and Jones 2003). To assess the efficiency of the value stream that composes the development process, the team can employ the element of metrics to investigate the efficiency of the process and its various activities and tasks (see Section 5.1.3).

The value stream should be carried out at several stages and it should include an element as quarterly value stream mapping to reassess the entire value stream process (Hibbs et al. 2009). This is an important consideration because it incorporates adding value to the various processes of the software development and ensures that the client is presented with a high-quality product (Poppendieck and Poppendieck 2003). Lean thinking has been argued by several scholars to be more concerned with doing the right work at the right time rather than concentrating on who is doing the right work (Bronza 2012).

5.1.2 Limit and Visualize WIP

Following visualizing, the workflow understands and puts limits on the work in progress (WIP). The WIP refers to the concurrent number of work items allowed in each process. The correct use of the Kanban board ensures minimal WIP, which highlights the constraints and teamwork coordination. This enables the team, as well as other stakeholders, to keep track of their development and distinguish between work items that are in progress and work items that are complete in the board. This visualization gives the team insight on how the development process is moving along.

Despite the fact that several studies ([S3][S5][S9][S12][15][S17][S18][S21][S26][S27][S29]) have explicitly identified limiting WIP as one of the core elements of the Kanban approach, only 13 percent of the primary studies reported their experiences on how to set WIP limits. The value of WIP has been outlined in almost 92 percent of the primary studies; for example, in [S6], Shinkle cited a statement that had been made by a developer during the Kanban implementation:

WIP limits seem to be the worst understood part of the kanban system. When used properly, it exposes bottlenecks and reduces lead time for individual work items. Used improperly, it can starve developers for work or result in too many people working on the same work items. [S6, p. 188]

In [S16], Oostvogel suggested that there should be set rules about the WIP limits along the development process. However, he stated, “*There is no exact science on determining WIP limits*” [S16, p. 39]. This fact among others has led more than 75 percent of primary studies to suggest constraining WIP limits by experiments. In [S5], Anderson established a general rule to define WIP limits: “*the work-in-progress limits should be agreed upon by consensus with up and down stream stakeholders and senior management*” [p. 113]. He argued that consensus creates commitment to adhering to the WIP limit.

A few studies endeavoured to establish formulas to generalize setting WIP limits based upon their own experiences. In [S5], for example, Anderson suggested that the WIP limit lies between 1 and 3 work items per person at any given time. Hence, if the team is composed of 10 members, and the consensus has been reached on having 2 work items per person, then the WIP limit is agreed to be 20 for the team. Other studies reported WIP as ranges; for example, in [S8] Terlecka reported the following:

The team reached the conclusion (later empirically checked by other teams) that the recommended limit of cards is between the number of team members plus 1, to number of team members times 2 minus 2, so for a 7-person team, it would be between 8 and 12. [S8, p. 100]

In [S15], Kniberg and Skarin suggested, “*The first WIP limit we used was $2n-1$ (n =number of team members, -1 to encourage cooperation). . . By monitoring the Kanban board, it is easy to figure out the right limits along the way*” [S15, p. 71]. They also provided three pages of example on limiting WIP (see [S15] pp. 19–22). Limiting WIP in [S6], [S31], and [S35] was suggested to be determined using Little’s law, which is one of the principles of queuing theory. Little’s equation is presented here:

$$WIP\ Limit = Average\ Processing\ Rate(Throughput) \times Cycle\ Time(Flow\ Time)$$

Several primary studies ([S10][S13][S33][S35][S37]) have discussed a combination between Little’s law and an innovative approach for controlling the flow and determining WIP limits. *Differential service* is an approach to differentiate quality of service by work stream. The approach suggests dividing the work items into categories with different classes of services. Then, the WIP limits can be set for each category (Reinertsen 2009); in that sense, the Kanban approach becomes more flexible and responsive to rapidly changing customer requirements.

The classification of work items helps prioritize the work activities, which leads to effective management and high- quality products. Work items are prioritized to avoid delays within the software development process. Because tasks are not equal, they are dealt with differently, which means that activities and tasks are treated according to their specific characteristics. The type of work determines the class given to the development process.

As suggested by [S3], the work items can be divided into categories of high, medium, and low categories. Another categorization was suggested by [S13], where work items were divided into the following class: “Normal Class of Service,” “Special Class of Service,” and “Expedite Class of Service.” Table 11 provides an example of how the combined approach of Little’s law and *differential service* can be used to specify the WIP limits for each category.

The example considers two developers who each one works 40 h per week; with the given throughput and the weekly time that is spent on each work item, the total working hours equals 80 h per week. The targeted cycle time per work item represents the time that is required for a work item to continuously flow throughout the development process without interruption. The WIP limit for each category of the service class is calculated using Little's formula by multiplying the throughput by the cycle time.

5.1.3 Measure the Flow

Measuring metrics is an important element in the Kanban approach. Based on the definition of the Kanban approach, the continual delivery of value to customers should be emphasized while promoting continuous learning and improvements. Hence, the primary goal of measuring flow is to pinpoint the improvement opportunities that will lead to a smoother flow to quickly deliver value to customers.

In measuring flow, it is important to remember that the system's capacity should not be exceeded. If the system is overloaded, the quality of the development will be low. WIP limits can be utilised to overcome the capacity overloading problems. The lower the WIP limits, the shorter the cycle time, presuming the throughput is constant, as there is a linear relationship between WIP and cycle time as shown in Little's law.

In [S5], Anderson expressed the relationships among WIP, lead time, and defects. He claimed that based upon industrial evidences, "*there is a correlation between increased lead time and poorer quality. Longer lead times seem to be associated with significantly poorer quality... Longer average lead times result from greater amounts of work-in-progress*" [p. 27]. Hence, measuring flow depends mainly on measuring the amount of WIP, lead time, and defect rates to gauge the quality of the product under development.

After identifying what should be measured and why it should be measured, the instruments to measure the flow can be narrowed down to the following tools: cumulative flow diagrams, lead time/cycle time, and defect rate. This is in line with what has been reported by about 30 percent of the primary studies ([S2][S3][S13][S15][S20][S24][S26][S27][S29][S33][S35]). Figure 9 provides an example of how the cumulative flow diagram is depicted and interpreted.

The figure shows three types of queues 1) Total, which represents the amount of work waiting in the queue to be pulled for development, 2) WIP, which represents the amount of work that is currently under development, and 3) completed, which represented the amount of work that was completed. The horizontal axis – X provides a good projection of the time required to complete the work item at hand, while, the vertical axis – Y, specifies the number of WIP.

Figure 9 represents a simple cumulative flow diagram followed by a naïve explanation; measuring the flow is a significant area to investigate and study; in fact, in [S24], Petersen and Wohlin produced a full research paper on the topic in which they empirically evaluated defined measures that had been applied to the visualized cumulative diagrams to increase throughput and reduce lead-time. Furthermore, Reinertsen ([S35]) devoted an entire book to the principles of product development flow.

Table 11 Example of Specifying WIP Using Little's Law and Differential Service

Category	Throughput per Week	Weekly Time per Work Item	Targeted Cycle Time per Work Item	WIP Limit
Expedite	3	2.5	1	3
Special	5	3.5	2	10
Normal	11	5	3	33

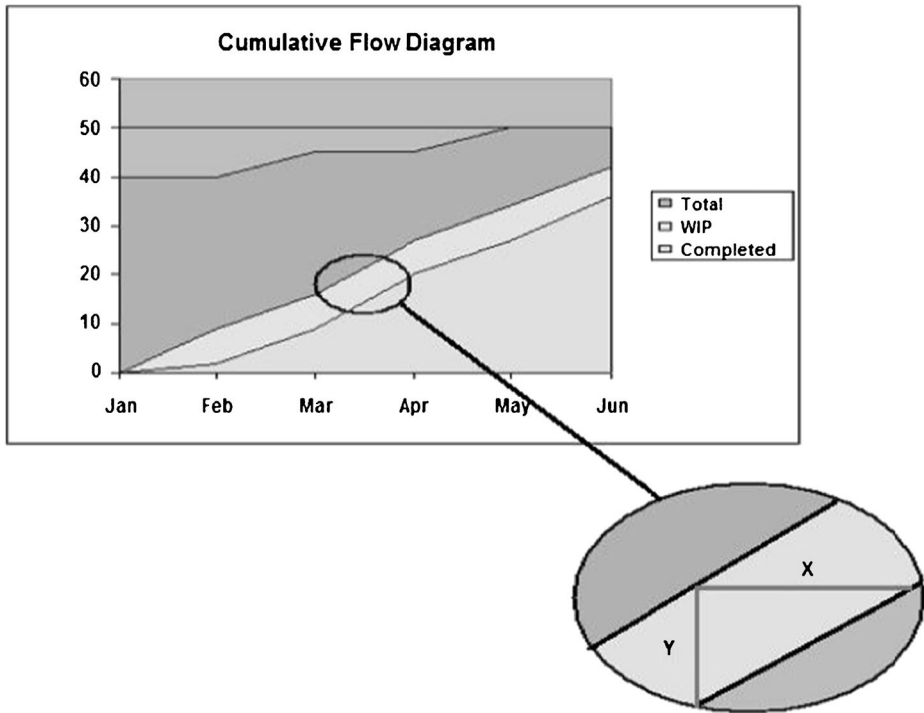


Fig. 9 Cumulative Flow Diagram

5.2 Validity Threats

An overall challenge experienced when undertaking this research study was the definition of the scope, as the area under study is multidisciplinary and covers a wide range of fields. Searches covering all the involved disciplines would be difficult and complex, as terminology may refer to a different notion in each field; this comprises a threat to *construct validity*. This threat was considered during the definition of the search criteria and has led to the avoidance of a high risk of errors. However, we could not discount the impact of confounding and local variables in having significant impact on the search results, which is considered a threat to *internal validity*.

Another limitation of this study forms a threat to *content validity*, as we had no control over the quality and level of details presented in the primary studies. Since most papers lacked adequate information for classifying them, it would have been preferred to have parallel data extraction and to *crosscheck* the information on all the study papers. Nevertheless, a lack of resources made this impossible.

Another threat is selection bias that compromises a *reliability threat*. To protect the study against this threat, the research took three strategies. Firstly, to balance precision and comprehensiveness of the search string, the trial searches were performed using Google Scholar where several alternative keywords were used and keywords were also combined. Secondly, the publications were selected from different sources such as the Scopus, Scirus, and InfoQ among others. Thirdly, measures were taken to avoid and limit built-in bias on the selection that was based on title, keywords, and abstract. It is possible that we may have missed relevant papers that would have provided great insight to the implementation of the lean-Kanban approach. This possible omission comprises a threat to *external validity*.

6 Conclusions

This systematic review incorporated 37 primary studies in which the context ranged from large to complex projects in various companies and also individual projects. The majority of the studies reported the implementation of lean practices in general, rather than focusing on the Kanban approach. The studies, however, reported the Kanban approach as the most frequently used lean practice. Further, a number of recently published studies (from 2006 onwards) illustrated a specific use of the Kanban approach, which is very much in line with the Wang et al. (2012) study. They noted the trend to use more specific lean practices, in particular the Kanban software development approach.

The Kanban approach to software development is a more direct implementation of lean product development. In software development, lean does not only refer to the lean manufacturing, but also refers to the lean elements translation to the software development application. Kanban offers a less prescriptive element as compared to agile methods and has become a popular extension to traditional agile methods (i.e. Scrum and XP) (Hibbs, Jewett, and Sullivan 2009).

The lean methodology, via the Kanban approach, can be argued to be the fastest growing product development approach in the past decade. However, despite the interest in the approach, there is neither a standard definition of the Kanban approach nor a clear definition of its elements for software development as indicated by 27 percent of the primary studies. This lack of definition causes a series of challenges to individuals who want to implement the Kanban approach, which has led to the need to address its elements.

The use of the Kanban approach comes with several challenges, as reported by more than 50 percent of the primary studies, the conspicuous challenge is related to a lack of details and absence of the guidelines on how the Kanban approach can be used in IT organizations, as well as the lack of the guiding principles on how the different Kanban elements can be introduced. In an attempt to address these challenge, this study provided an analysis of the included 37 studies to investigate the reported Kanban approach's elements, uses, and presentations.

As derived from the available literature, more than 32 percent of the primary studies agreed on that the Kanban approach has three prime principles 1) Visualize the workflow, 2) Set WIP limits, and 3) Measure the flow. It is important for the development team to have consensus on how these three principles should be implemented and used. Some guiding principles have been synthesized from the primary studies, where they were used and proven successful. Team cohesiveness and teambuilding are important elements for ensuring a successful Kanban approach.

Software development teams in the various firms in the IT industry are considered to still operate as islands and they should instead work together to ensure system quality and consistency (Pink 2009; Oostvogels 2012). More than 16 percent of the primary studies reported that the use of the Kanban approach drops the use of the isolated teams and led teams to work together to ensure high-level quality as well limited use of the available resources. The team members should participate in making the Kanban board to ensure that there is a feeling of ownership and pride.

Despite the reported challenges, the Kanban board, as reported by almost 46 percent of the studies, is an efficient visualization tool and should be used effectively to ensure that the development process takes place as expected. This is not the only benefit of the Kanban approach, our findings show that more than 32 percent the primary studies used the Kanban approach to effectively deriving and leading organizational changes as well as facilitating the cross-functional teamwork.

The recommendation of this paper is to discuss the Kanban elements together based on the five pillars of the lean approach to minimize the risk of evolving contradictory elements and to reduce the gap in the literature by creating a common ground to define these elements. In turn, this facilitates establishing guidelines and instructions on how to setup the Kanban approach to give practitioners an overall framework that increases the likelihood of successfully implementing the Kanban approach in IT organizations.

The paper concludes that there is limited research that gives guidelines to the practitioners of the software development in implementing the Kanban approach. It is also evident that there is a strong need for systematic studies on the benefits of the lean-Kanban approach. This includes the analysis of what needs to be changed or improved within the lean-Kanban practices to their application in the IT industry. Further research is also needed to provide formulas to help practitioners in defining the primary elements of Kanban, such as inclusion criteria and defining the “done” items.

The correct use of the Kanban approach ensures that there are realized benefits to IT organization, including but not limited to, minimal work in progress; give a highlight on the constraints and a coordination of the teamwork. The lean approach is more than the Kanban method and it has been proposed that the Kanban approach should use more approaches from the lean principle so as to take the full advantage of the lean principles in the software development. The Kanban approach shares principles of the agile methodologies on the basis that the requirements are explained in atomic features, which are then implemented incrementally.

Appendix 1: List of Included Studies

[S1] Wang, X., Conboy, K., & Cawley, O. (2012). “Leagile” software development: An experience report analysis of the application of lean approaches in agile software development. *Journal of Systems and Software*, 85(6), 1287–1299.

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[S5] Anderson, D. J. (2003). *Agile management for software engineering: Applying the theory of constraints for business results*. Prentice Hall Professional.

[S6] Shinkle, C. M., & Shihkle, C. M. (2009). Applying the Dreyfus Model of Skill Acquisition to the Adoption of Kanban Systems at Software Engineering Professionals (SEP). In *Agile Conference, 2009. AGILE'09*. (pp. 186–191). IEEE.

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[S8] Terlecka, K. (2012). Combining Kanban and Scrum – Lessons from a Team of Sysadmins. In *Agile Conference (AGILE), 2012* (pp. 99–102).

[S9] Nikitina, N., & Kajko-Mattsson, M. (2011). Developer-driven big-bang process transition from Scrum to Kanban. In *Proceedings - International Conference on Software Engineering* (pp. 159–168).

[S10] Turner, Richard, Ingold, D., Lane, J. A., Madachy, R., & Anderson, D. (2012). Effectiveness of kanban approaches in systems engineering within rapid response environments. *Procedia Computer Science*, 8(0), 309–314.

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- [S19] Anderson, D. J., & Garber, R. (2007). *A Kanban System for Sustaining Engineering on Software Systems*. Retrieved April, 01, 2013 from <http://www.lean.org/FuseTalk/Forum/Attachments/Kanban%20for%20Software%20Development-Corbis.pdf>
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Appendix 2: Quality Instrument

		Quality Instrument		
Study Identifier (Unique ID)				
1	What is the study Type?	☐ ML	☐ SLR	☐ GL
<i>If the type is Monographic, complete section (A), if the type is SLR, complete section (B), otherwise, complete section (C)</i>				
Section (A): Monographic				
1	Does the study report any empirical research?	☐ Yes	☐ No	
2	Is there a clear description of the aim of the research?	☐ Yes	☐ No	☐ Partially
3	Is there a clear description of the study context?	☐ Yes	☐ No	☐ Partially
4	Is there an adequate justification for the research design?	☐ Yes	☐ No	☐ Partially
5	Is there an adequate description of the studied sample?	☐ Yes	☐ No	☐ Partially
6	Is there an adequate justification for the selected sample?	☐ Yes	☐ No	☐ Partially
7	Is there a clear description of the data collection?	☐ Yes	☐ No	☐ Partially
8	Is there a clear description of the data analysis?	☐ Yes	☐ No	☐ Partially
9	Are the findings of the study clearly stated?	☐ Yes	☐ No	☐ Partially
10	Is there sufficient data presented to support the findings?	☐ Yes	☐ No	☐ Partially
11	Are the research contribution and value adequately discussed?	☐ Yes	☐ No	☐ Partially
12	Are validity threats of the research adequately described?	☐ Yes	☐ No	☐ Partially
<i>If any of the answer is "No" for any of the first two questions, exclude the paper. The scoring procedure is Yes=2, Partial=1 and No or Unknown=0, calculate total from 4-12, if score is 12 or more, include. For more details about scoring, see "Ivarsson and Gorschek".</i>				
Section (B): Systematic Literature Review (SLR)				
1	Are the review's inclusion and exclusion criteria described and appropriate?	☐ Yes	☐ No	☐ Partially
2	Is the literature search likely to have covered all relevant studies?	☐ Yes	☐ No	☐ Partially
3	Did the reviewers assess the quality/validity of the included studies?	☐ Yes	☐ No	☐ Partially
4	Were the basic data/studies adequately described?	☐ Yes	☐ No	☐ Partially
<i>The scoring procedure is Yes=2, Partial=1 and No or Unknown=0, calculate total from 1-4, if score is 6 or more, include. For more details about scoring see "Kitchenham - Guidelines".</i>				
Section (C): Gray Literature				
Websites				
1	Is it an official website for Lean/Kanban/Agile/Software organizations/societies?	☐ Yes	☐ No	
2	Does it provide references for what has been written/described on the website?	☐ Yes	☐ No	
3	Is the data on the website quoted/cited from one or more of the included primary	☐ Yes	☐ No	
<i>If any of the answer is "No" for any of the first two questions or "Yes" for the third question, exclude the website.</i>				
Non-Websites (eBooks, Articles on InfoQ)				
1	Does the study report any empirical research?	☐ Yes	☐ No	
2	Has the study been peer reviewed by an expert in the domain?	☐ Yes	☐ No	
3	Does the study have "Known" author?	☐ Yes	☐ No	
4	Does the author has h-index > 5? (using Google Scholar gadget)	☐ Yes	☐ No	
5	Does the study has been cited by > 5 studies? (using Google Scholar gadget)	☐ Yes	☐ No	
<i>If the answer is "No" for the 1st questions, exclude the study. If the answer is yes, proceed to question 2, if the answer is "Yes", include the study, if the answer is "No", proceed to question 3 and 4, if the answer is "No" for both of the questions, exclude the study.</i>				

Appendix 3: List of Excluded Studies

Akbar, R., Hassan, M. F., & Abdullah, A. (2011). A Review of Prominent Work on Agile Processes Software Process Improvement and Process Tailoring Practices. In *Software Engineering And Computer Systems, PT 3* (Vol. 181, pp. 571–585).

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Cockburn, A. (2006). *Agile software development: the cooperative game*.

Corona, E., & Pani, F. (2012). An investigation of approaches to set up a Kanban board, and of tools to manage it. In *SITE'12 Proceedings of the 11th international conference on Telecommunications and Informatics, Proceedings of the 11th international conference on Signal Processing*, pp. 53–58.

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Sanders, A. (2007). Kanban Ground Rules Example for a Specific Team. Retrieved April 01, 2013, from <http://aaron.sanders.name/agile-fashion/kanban-ground-rules-example-for-a-specific-team>

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Appendix 4: Coding scheme

Category	Description	Coding Rule
Kanban method	It refers to the method of incremental and evolutionary process change for organizations and institutions.	Any description of the steps that has been undertaken to implement the Kanban approach into an organization.
Kanban Board	This refers to visual tools in project management application that have a basis on Kanban for the visualization and optimization of the workflow. It also enables real time teamwork and team collaboration	Any tool or means used to visualize the processes, tasks, or the workflow of the software product development.
Pull System	This refers to trigger the process of working on producing only what the customer needs by the quantity that is needed and only when it is needed	The approach of starting the development process based upon a request from the customer.
Work Item	It refers to a task that needs to be visualized and added to the Kanban board.	Any feature, user stories, MMF or any other item that needs to be visualized and has to go through different phases through the development process.
Priority Queue	It refers to the list of ordered requirements or work items	Any term indicates the use of queuing, ordering, or prioritizing tool. i.e. backlog or to-do items.
Inclusion Criteria	It refers to the rules that are used to make the decision of adding the work item to the visualization tool. Ideally, it should consider the value from a customer perspective.	Any norm, rule, or standard that is used to determine what work-item should be added to the board, i.e. through the use of MMF or Story point.
Work In Progress (WIP)	It refers to the concurrent number of work items that is allowed in each development phase.	Any rule, technique, to determine the number of concurrent work items.
Done Items	These are the items that have been completed on a particular phase and are ready to be released to the next stage.	The work items that have been through a specific development process and ready to be pulled by the next development process.
Reverse Items	It refers to items that are being moved backwards on the Kanban board instead of moving forward.	Any work item that needs to go back to the previous development process, i.e. bug fixing.
Validated Learning	This variable refers to the process of learning what works and what does not work for a specific organization. Its intent is to measure the a work item's value from a business perspective	Validate learning is whereby an organization over time obtains a formula for obtaining, qualifying, and selling to customers in a particular target area. This can refer to the P-D-C-A cycle.
Cycle Time	Cycle time is the actual time required to complete one cycle of an operation for one work item from start to finish while the work item flows seamlessly through the development process.	Cycle time is a calculated based upon the time spent on producing a work item from the beginning until delivery without delays.
Lead Time	This is the time of completing one work item including delays between the initiation of a process and its execution, i.e. the time between making an order and the order being delivered.	Lead-time is calculated from the moment of requesting the work item until the moment that item is delivered to the customer.

Category	Description	Coding Rule
Measurement tool	This usually a diagraming tool illustrates the cycle taken by a work item as it goes through the system.	Any measurement diagram is used to represent the performance of the Kanban.
Bottleneck	Bottlenecks are the constraints that are faced during the product development process.	Any work item that is slacking and slowing the process down.
Slack / Buffer	This refers to designed quantity of time applied to task schedule to protect the success implementation of the required deliverable on time.	Any spare time put aside to help in delivering the work item on time.
Kanban Principles	These principles offer a framework that is chosen in the software development process by using the Kanban approach.	These are the guiding concepts of the Kanban approach.
Kanban techniques	This refers to production control system that is used to overcome the problems once occurred.	Any technique used to allow no surplus production and gives the workers the right to stop if they cannot keep up.
Avatar	A visual representation of the work item details.	Any visual representation of the details related to a work item, this includes sticky notes, symbols or any other means.

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